

E9 205 – Machine Learning for Signal Processing

Homework # 2 - Part B

Due date: Oct 26, 2016 (3:30PM)

October 17, 2016

Data for second assignment is posted here

<http://leap.ee.iisc.ac.in/sriram/teaching/MLSP/assignments/HW2/Data.tar.gz>

1. **Fischer faces** - 15 subject faces with happy/sad emotion are provided in the data. Each image is of 100x100 matrix. Perform PCA on to reduce the dimension from 10000 to K and then perform LDA to one dimension. Plot the one dimension features for each image. Select the optimum threshold to classify the emotion and report the classification accuracy on the test data. What is the optimum value of the K ? **(Points 15)**

2. **Implementing GMM** - A set of training and test examples of music and speech are provided.

Using these examples,

- a Generate mel spectrogram features - Use the spectrogram as before and apply the Mel filters as windows FFT and integrate. For 512 component magnitude FFT, mel filters are here www.ee.iisc.ernet.in/new/people/faculty/prasantg/downloads/melFilters.mat (currently in Matlab format - will provide the ascii format.). After mel integration, logarithm is applied to get mel spectrogram features ($40 \times N$, where N is the number of frames).
- b Train two GMM models with K-means initialization (for each class) separately (i) 2 mixtures with diagonal covariance, (ii) 2 mixtures with full covariance and (iii) 5-mixture components with diagonal/full covariance respectively on this data.
- c Classify the test samples using the built classifiers and report the performance in terms of error rate (percentage of mis-classified samples) on the test data.
- d What is the impact on the performance for different number of mixture components, diagonal versus full covariance ?

(Points 20)

3. **Implementing HMM** - Use the same data and mel spectrogram features generated in the last question.

- a Write a code to implement the likelihood computation using the forward variable after assuming a uniform flat start based initialization with 3 and 5 states per HMM and GMM with 2 mixture components per state.

- b Write a code to implement the Viterbi algorithm to decode the best state sequence using the existing model.
- c Use the Baum-Welch reestimation method to train HMMs with examples from music and speech features.
- d Classify the test examples and report the performance. How does the performance change for different number of states per HMM, different number of mixture components per GMM ? Will diagonal or full covariance GMM be a good choice ?

(Points 25)